Timothy Do

San Jose, CA | +1 (408) 644-5538 | timothydobsa@gmail.com | linkedin.com/in/do-timothy | github.com/dotimothy

Education

University of California, Irvine — Irvine, CA
Bachelor of Science in Electrical Engineering, Accelerated Status — *Sep. 2019 – June 2023*

- **Cumulative GPA:** 3.903/4.0, **Major GPA:** 3.91/4.0
- **Specialization:** Digital Signal Processing
- **Clubs and Societies:** IEEE, IPC, Eta Kappa Nu, Tau Beta Pi

Experience

Undergraduate Student Researcher — January 2022 – Present
Irvine Computer Vision Laboratory at UC Irvine — *Irvine, CA*

- Assisted in research under the direction of Professor Glenn Healey on the topic of Spectral Image Filtering for Emissions on Wildfires
- Digitized Imaging Filters into CSVs and compared them in MATLAB against certain chemical emissions.
- Generated altitude maps using QGIS & SRTM 1 Arc-Second Global

Industrial Internet of Things Intern — June 2022 – Sep. 2022
Western Digital — *San Jose, CA*

- **Skills:** MySQL, Python, Raspberry Pi, Soldering, Linux, Lean Six Sigma
- Certified as a Six Sigma Yellow Belt for learning Six Sigma foundations
- Developed Presentation Skills in a Corporate Environment.
- Applied Industrial Tool Sets and Embedded System Programming on Sensor Data Analytics

Vice President — Nov. 2021 – Present
Institute of Printed Circuits Club at UC Irvine — *Irvine, CA*

- Reserved club meeting rooms and sent weekly emails to club members regarding club events
- Programmed the club website and managed social media (i.e. moderator of club discord server)
- Led a Two-Part AM Radio Workshop where the circuit was designed for the best amplifier gain in addition to creating easy-to-read slides & ordering parts from Amazon/Dollar Tree.

Projects

PhotoLabServer | *Python, Matlab* — August 2022

- Implemented a Python Flask Web Server to process images
- Piped with HTTP Post Requests to send images & back with HTTP GET
- Performed image operations like sharpen, smoothing, FFT using MATLAB/Python backend

EECS195-Colorization | *Python, MATLAB* — Feb. 2022 – March 2022

- Colorized grayscale images using GAN and Auto-Encoder Neural Networks
- Gathered own datasets by taking pictures of different scenery
- Clustered pixels into superpixel subgraphs for uniform training weights

TTLZNetChess | *C, Makefile, git* — March 2021 – June 2021

- Managed a team of four developers to develop a C-Based Chess Application
- Implemented Terminal Graphics (Colors) using Escape Codes
- Implemented AIs in Chess using Minimax Algorithm
- Added Networking Support using standard sockets and FCFS Protocol

PiFmMorse | *Python, bash* — Septemeber 2020

- Programmed a Raspberry Pi to transmit Morse Code with FM over a GPIO Pin
- Concatenated .wav sound files to parse Morse Code messages to transmit
- Called Linux programs to interface with GPIO Pins

Skills

Electronics: Computer Assembly, Cadence, LTSpice, Soldering, Oscilloscope & Multimeter Measurements
Programming Languages: Python, Java, C, CUDA, MATLAB, bash, MySQL HTML/CSS/Javascript, VHDL
Developer Tools: Git, SSH, Makefile, Docker, Azure, VS Code, Visual Studio, PyCharm, Eclipse
Libraries: pandas, NumPy, Matplotlib, OpenCV, tensorflow, pytorch